



The Energy Cost of Not Acting Now:

A Case Study on Google Data Centers

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1. Background

Data centers are playing a more critical role in powering the digital world. However, it comes at a high environmental cost due to the massive energy consumption. Among them, Google's data centers are an ideal case to study for their outstanding scale and sustainability efforts. This report studies Google's current path, evaluates the gaps and examines the risks of delayed action on sustainability, focusing on the energy costs and the climate impacts associated with inaction.

2. Introduction

2.1. Client Overview

Google is a global technology leader, famous for its innovation and massive digital infrastructure. From search engines to cloud computing and AI, Google's services rely on powerful data centers which consume huge amounts of energy and contribute significantly to the environmental footprint of the company. Recognizing these challenges, Google has made sustainability a priority in its data centers' operations.

2.2. Scope of Analysis

This report assesses the energy efficiency and climate impact of Google's data centers through the analysis of its energy consumption, carbon emissions and sustainability practices. It predictively analyses the risks of delayed action and explores strategies for a sustainable transition.

2.3. Objective

The objective is to evaluate the energy cost of inaction, identify associated risks, and provide actionable recommendations to address these challenges effectively.

3. Identifying Current Actions and Data Sources

3.1. Current Sustainability Actions

Google has taken several actions to enhance sustainability of its data centers, including:

- **Renewable energy integration:** signing power purchase agreements (PPAs) to source renewable energy, achieving a global carbon-free energy (CFE) score of 67% in 2023.
- **Carbon offset purchases:** investing in offset programs to neutralize residual emissions.
- **Carbon capture and storage:** exploring emerging technologies to sequester carbon.

- **Optimized cooling systems:** adopting optimized cooling methods to minimize water and energy usage.
- **Water conservation strategies:** optimizing water use and implementing alternative cooling methods in water scarce regions.

Taking Google's Eemshaven Data Center in the Netherlands [1] as an example, it exemplifies its commitment to sustainable operations. To reduce freshwater usage, the facility utilized a €45 million cooling system that sources up to 10 million cubic meters of industrial water annually through a 28-kilometer pipeline connected to Groningen's industrial park [2]. This system relies on treated surface water and operates in a sustainable, carbon-free manner, significantly reducing the environmental impact of its cooling processes.

In addition to water sustainability, Google integrates renewable energy sources into the data center's operations. Collaborating with Eneco [3], the facility is partially powered by wind energy, demonstrating Google's effort to achieve 24/7 CFE operations. These measures highlight Google's innovative approach to addressing environmental challenges while maintaining operational efficiency.

3.2. Data Sources

The analysis is based on diverse data sources, including Google's environmental disclosures [4][5], third-party energy and carbon data, climate models and projections.

4. Scenario Analysis: Impact of Not Acting Now

4.1. Scenario 1: Google's Current Path

4.1.1. Internal and External Challenges

Google's data centers face a combination of internal and external challenges that put pressure on their ability to keep sustainability. These challenges, led by global warming, rising demand of Artificial Intelligence (AI) and increasing greenhouse gas (GHG) emissions, are expected to increase Google's future energy costs. Without imminent measures, these trends could deviate Google's commitment to renewable and climate-friendly policies.

(1) Global Warming and Temperature Increases

As temperature rises, the need for cooling systems grows. Meanwhile, extreme weather events like droughts and storms make operations less efficient and increase maintenance costs. In regions like Europe and North America, droughts are becoming more common, which reduces the availability of freshwater for cooling [6]. Google's data centers use around 20 billion liters of drinkable water—equivalent to the yearly consumption of 600,000 people.

To mitigate this, Google aims to replenish 120% of its water use by 2030. But by 2023, it has managed only 20% [7]. Oppositely, increased rainfall causes flooding and water contamination, raising treatment costs and affecting cooling system efficiency.

(2) Rising Demand for AI

As AI is becoming an essential part of daily life, its computational demand is growing exponentially. Consequently, it contributes to higher energy consumption and carbon emissions. While it's hard to pinpoint exactly how much AI contributes to global emissions, it is estimated that it currently covers 2-3% of the total [8]. To put it in perspective, a single ChatGPT search uses about 2.9 watt-hours of electricity, compared to just 0.3 watt-hours for a Google search [9]. Referring to the International Energy Agency (IEA) [10], it's projected that in 2026 the global demand for data centers, AI and cryptocurrencies will be 1000 TWh, more than twice the value in 2019 of 460 TWh. To address this, Google invested \$1 billion in a new data center in the UK [11].

AI also presents challenges related to water usage. By 2027, the sector is expected to use between 4.2 and 6.6 billion cubic meters of water [12], roughly half of the United Kingdom's annual water consumption [13]. In 2022, Meta, Microsoft, and Google collectively consumed 2.2 billion cubic meters of water, further amplifying AI's environmental impact.

(3) Increasing GHG Emissions

Energy consumption and GHG emissions go hand in hand. While Google is pushing to reach net zero emissions by 2030, its emissions have continued to rise. According to Google's 2024 environmental report, emissions increased by 13% in 2023, nearly 50% more than in 2019 [5]. This increase is largely driven by higher energy consumption and supply chain emissions.

In addition to environmental concerns, Google will face economic risks if emissions are not adequately controlled. The introduction of carbon taxes and environmental penalties in many countries could increase operational costs. Not to forget that Google's target of net-zero carbon emissions by 2050 is in jeopardy, as currently the energy consumption trend overcomes its sustainability efforts.

4.1.2. Reported Impacts and Gap Analysis

This section evaluates Google's historical performance in key sustainability areas, and identifies trends, challenges and gaps relative to its 2030 goals.

(1) Green House Gas (GHG) Emissions

From 2019 to 2023, Google's total GHG emissions show an increasing trend with a

compound annual growth rate (CAGR)¹ of 10.30%. The claimed total emissions are much lower than the real total emissions because the part offset through Power Purchase Agreements (PPAs) is considered.

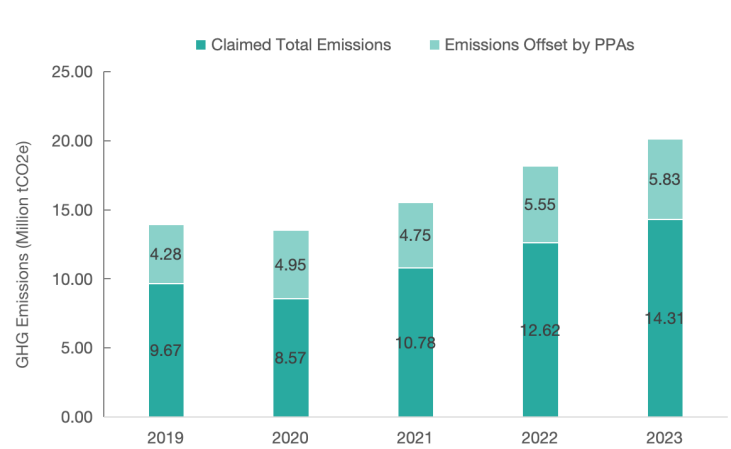


Figure 1: GHG Emissions of Google (Data Centers, Offices, Value Chain)

The carbon intensity of energy consumed (including PPAs offset) also presents an increasing trend from 2019 to 2023 with a CAGR of 17.18%. It indicates external factors and challenges related to global energy systems and renewable integration, although Google has put huge investments in renewable energy, energy storage, and AI optimization.

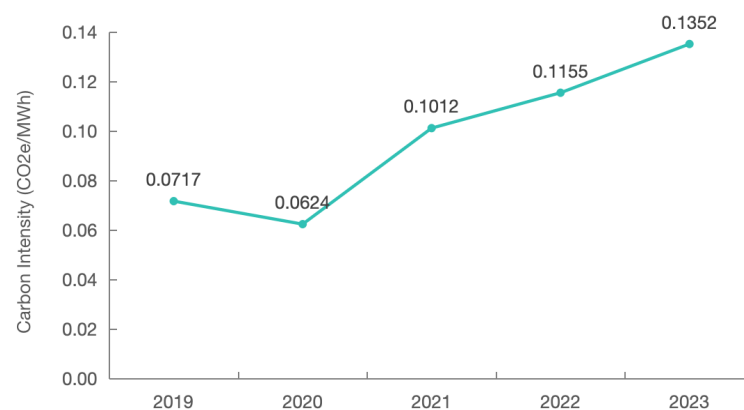


Figure 2: Carbon Intensity of Energy Consumed (Data Centers, Offices)

(2) Energy Consumption

Taking the energy consumption structure of year 2023 as an example, Google's total energy consumption includes all fuel and natural gas consumption; purchased electricity, steam, heating, cooling; and all electricity generated on-site from renewable sources. Among them, purchased electricity accounts for 98%.

¹ CAGR represents the average annual growth rate over a period time, assuming consistent yearly growth that, in this case, highlights a steady and significant increase in emissions during the study period.

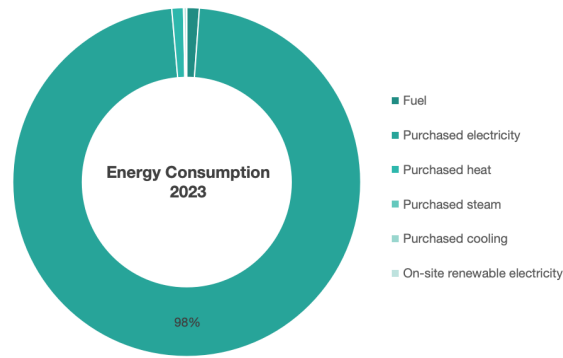


Figure 3: Energy Consumption Structure of Google's Data Centers

From 2019 to 2023, the purchased electricity increases from 12.24 TWh to 25.31 TWh, with a CAGR of 19.92%. Google claims a 100% renewable energy matching on a global and annual basis. However, the structure analysis of renewable sources shows that more than 75% of renewable electricity comes from PPAs, over 20% comes from local grid and only around 0.05% comes from on-site renewable energy projects.

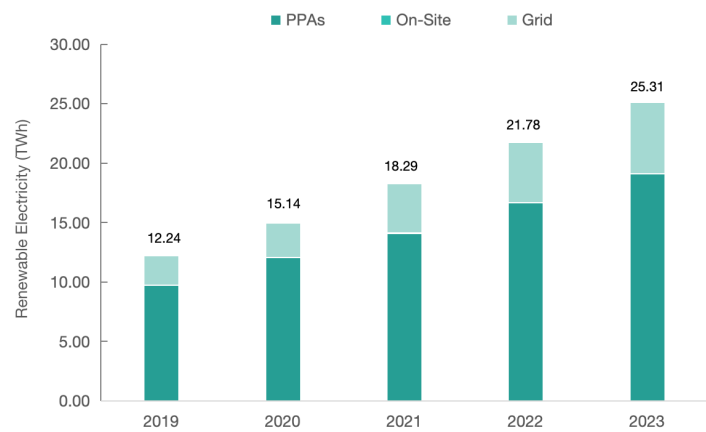


Figure 4: Electricity Purchased from Renewable Sources

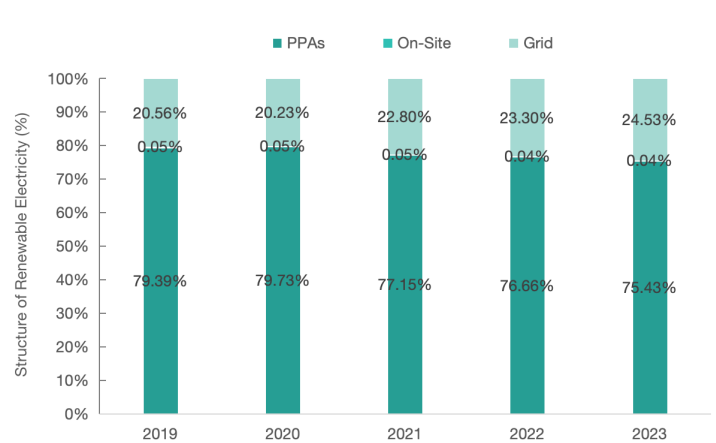


Figure 5: Structure of Purchased Renewable Electricity

(3) Carbon-Free Energy (CFE)

In comparison with annually matching the energy use with renewable energy purchase, it's far more complex and technically challenging to run on 24/7 carbon-free energy on every grid. Taking year 2023 as an example, Google's global average CFE score for its data centers is around 67%, and it shows obvious regional variations. Regions like North America – Central and Latin America have a CFE of 91%, while the CFE score of Asia Pacific is only 12%. These regional variations indicate uneven access to clean energy, which is a potential risk for its sustainability goal. Google needs to purchase additional renewable energy credits (RECs) or offsets for regions with low CFE, which doesn't truly eliminate local carbon emissions. For example, by procuring larger amounts of wind energy in places like Europe, Google compensates for its lack of renewable energy purchases in the Asia-Pacific region.

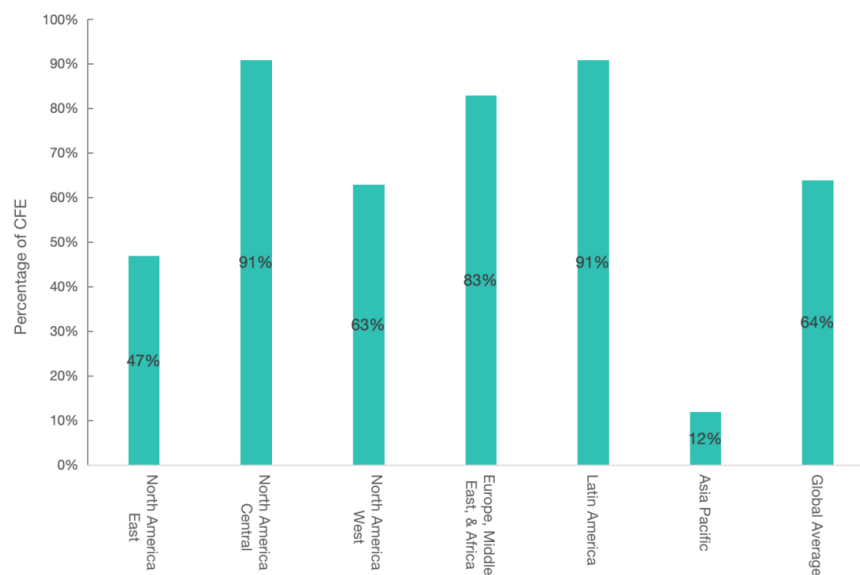


Figure 6: Regional Average CFE across Google's Data Centers (Hourly, 2023)

(4) Gap analysis

Google has set a goal to achieve net-zero emissions across all their operations and value chain by 2030. In terms of carbon reduction, the target is to reduce 50% of their combined Scope 1, Scope 2 (market-based), and Scope 3 absolute emissions (compared to 2019 base year) by 2030. Using the CAGR of GHG Emissions (market based) from 2019 to 2023 as forecast reference, the GHG Emissions in 2030 is estimated to be 28.43 Mt CO₂, which indicates a huge gap of 23.59 Mt CO₂. The gap is caused by increases in data center energy consumption and supply chain emissions.

In terms of carbon free energy, the target is to run on 24/7 carbon-free energy on every grid

where they operate by 2030. If Google follows the increasing rate in the past five years, the carbon free energy is estimated to be 70% in 2030, which shows a 30% gap compared with its goal.

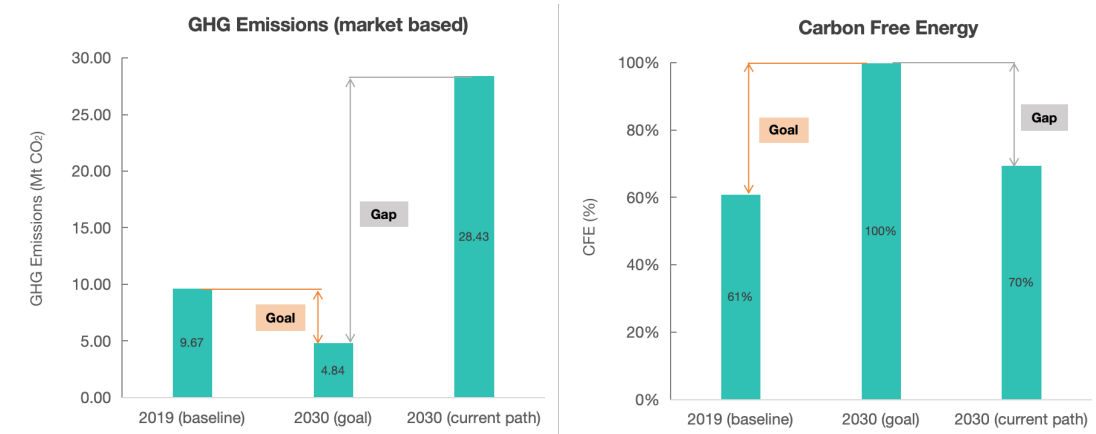


Figure 7: GAP Analysis of Google's Goal and Current Path

4.2. Scenario 2: Optimized Path

4.2.1. Predictive Analysis of Energy Cost

Considering the broader context of global warming and the future energy demands driven by AI workloads and data center expansion, this chapter provides a detailed predictive analysis of energy costs for data centers in 2030, focusing on the impacts of carbon emissions, energy consumption growth, and global warming.

(1) Carbon Emissions

According to the previous calculation, the GHG Emissions in 2030 is estimated as 28.43 Mt CO₂, which is the remaining emissions after PPAs offset. Carbon policies are expected to become significantly stricter by 2030 as countries and regions aim to meet their climate goals under the Paris Agreement. Taking the forecast carbon price in Europe [14] as reference, the cost of remaining carbon emissions is calculated as follows.

Table 1: Carbon Cost in 2030

Items	Carbon Price (€/t CO ₂)	Carbon Cost (Million €)
Highest Forecast	157	4463.78
Average Forecast	140	3980.44
Lowest Forecast	78	2217.67

By 2030, failing to adapt to these stricter policies will likely result in significant financial and operational challenges for Google's data centers.

(2) Energy Consumption

The rapid demand of AI applications and training models and expansion of the data center sector is posing an increased energy demand challenge to the grid. According to Goldman Sachs Research [9], global power consumption of data centers in 2023 is 411 TWh and is estimated to be 1063 TWh in 2030, with a CAGR of 14.54%. This boom in energy consumption will require extra investment in grid utilities especially in areas where the power grid is old.

Using this increasing rate as a reference, Google's electricity consumption in 2030 is calculated as 65.45 TWh, which is around 260% of that in 2023. Taking Europe energy price as an example, more renewables, electrification and flexibility can slash average day-ahead energy prices by 25% by 2030, estimated by SolarPower Europe [15]. Hence, the electricity cost for google in 2030 will be around 195% of that in 2023, which is also a huge increase.

(3) Global warming

Studies indicate that for each 1°C increase in ambient temperature, data center cooling systems may consume up to 4% more energy [16]. As energy consumption by cooling data center IT equipment can be over 40% of total energy consumption [17], the energy cost of global warming could be high for data centers. Even if the temperature increase caused by global warming can be controlled at 1.5°C by 2030, the extra energy consumption cost by global warming is calculated as 1.57 TWh. Using the 2023 IEA median industrial electricity price €0.2137/kWh [18] as the reference and considering the 25% electricity price drop in 2030, this extra energy consumption represents a 251.8 million € extra energy cost for Google.

Meanwhile, extreme weather conditions driven by global warming are expected to have significant impacts on data centers. The frequency of power outages and grid instability due to storms or extreme temperatures requires additional investments in backup power systems. Moreover, the physical risks of flooding or storm damage may necessitate costly infrastructure reinforcements. These factors, alongside the need for more energy-efficient systems, contribute to rising operational costs for data centers.

4.2.2. Proposal for Google: Optimized Path and Solutions

To address the escalating energy cost challenges and align with its sustainability goals, Google can take the following strategies into consideration.

(1) Enhance Carbon Free Energy Utilization

To achieve the 24/7 CFE goal across all regions of its data centers, Google should prioritize investments in carbon-free energy projects, particularly in areas with low renewable energy penetration, such as the Asia-Pacific region, which typically has lower CFE scores. Rather than relying on additional RECs from other regions, Google can create a more significant impact by partnering with governments and local utilities to improve access to renewable energy and boost local renewable energy penetration.

(2) Increase On-Site Renewable Energy Generation

To reduce dependence on PPAs and grid-based renewable energy, it is recommended to install solar panels and wind turbines directly at data center locations. Combining these on-site systems with energy storage solutions like battery energy storage systems (BESS) can significantly improve the efficiency and reliability of renewable energy use.

(3) Apply Advanced Cooling Technologies

Considering the increased cooling demand due to global warming, it's very important to apply advanced cooling technologies to improve efficiency while lower down the cost. Liquid cooling technology, which includes cold plate, immersion, and spray cooling, is one of the more promising cooling methods for data centers [19]. Shifting to liquid cooling systems can be a better option than traditional air-based cooling. Moreover, AI-driven tools can also be applied to dynamically monitor and adjust cooling systems for energy efficiency without compromising operational performance.

(4) Develop Climate-Resilient Approaches

To address the growing risks of extreme weather, building flood-resistant and storm-hardened facilities, and implementing modular designs that can adapt to changing environmental conditions are recommended as direct climate-resilient approaches. Moreover, creating a closed-loop system that reuses heat generated by data centers and wastewater from cooling systems can be particularly beneficial during extreme weather conditions that disrupt traditional energy supplies.

By implementing these solutions, Google can achieve sustainable growth, reduce financial risks associated with carbon and energy costs, and contribute positively to global carbon reduction efforts.

5. Conclusions and Open Questions

5.1. Summary of Findings

If Google follows its current sustainability efforts, meeting its 203 GHG emissions reduction and 24/7 CFE targets will be extremely difficult. The energy cost of inaction is high, with rising energy consumption, GHG emissions, and operational disruptions posing significant risks to Google's sustainability goals. However, with targeted investments in renewable energy, advanced technologies, and climate resilience, these risks can be managed while enhancing operational efficiency.

5.2. Recommendations

To address energy cost challenges and support sustainability goals, Google should focus on the following key strategies:

- **Enhance Carbon-Free Energy Use:** Invest in renewable energy projects, especially in regions with low renewable energy access, and collaborate with local governments and utilities to improve energy availability.
- **Increase On-Site Renewable Energy Generation:** Expand the use of solar and wind energy at the sites of data centers and integrate with energy storage systems to enhance efficiency and reliability.
- **Adopt Advanced Cooling Technologies:** Use more efficient cooling solutions, such as liquid cooling technology and AI-driven systems, to optimize energy use and reduce cooling-related costs.
- **Develop Climate-Resilient Infrastructure:** Design adaptable infrastructure that can handle extreme weather conditions and incorporate systems to reuse heat and water to ensure sustainability during environmental disruptions.

5.3. Open Questions

There are some unsolved questions for further research.

Our predictive analysis of energy cost is based on simplified model without considering regional differences. Lots of parameters like the weather conditions, grid composition, GHG emissions, carbon price and electricity price vary in different regions. If we make deep study in the future, a more complex model should be set up.

Most of the data is from Google's disclosure. We notice that it's difficult for Google to address its supply chain emissions accurately, which means the real emission and energy cost are not easy to reach.

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Declaration of generative AI and AI-assisted technologies

During the preparation of this work, we used ChatGPT to help us to collect data and find related reports and websites more efficiently. Also, we partially use ChatGPT to improve the language of the report. After using this tool, we reviewed and edited the content as needed and take full responsibility for the content we have submitted.